Neural Layer Matrices and Activations

ANSWER KEY

Section 1: Affine Transformations and Matrix Dimensions

1 A fully-connected neural layer maps input $\mathbf{x} \in \mathbf{R}^{128}$ to output $\mathbf{y} \in \mathbf{R}^{64}$ via $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. What is the shape of the weight matrix $\mathbf{W}$?

a. 64×128 ✓

b. 128×64

c. 128×128

d. 64×64

To multiply a (128×1) input and yield a (64×1) output, $\mathbf{W}$ must have shape 64×128.

2 For a mini-batch of $B = 32$ samples with input dimension $D = 128$, what is the shape of the activation batch $\mathbf{Y} = \mathbf{X}\mathbf{W}^T$ when $\mathbf{W}$ is 64×128?

a. 32×64 ✓

b. 64×32

c. 32×128

d. 128×64

$\mathbf{X}$ is (32×128), $\mathbf{W}^T$ is (128×64). Product shape: 32×64.

3 If a two-layer linear network computes $\mathbf{y} = \mathbf{W}_2(\mathbf{W}_1\mathbf{x})$ without non-linear activations, why is its expressive capacity limited to a single linear transformation?

a. **The product of two matrices $\mathbf{W}_3 = \mathbf{W}_2\mathbf{W}_1$ is itself a single matrix** ✓

b. The gradient of two matrices is always undefined

c. Matrix multiplication is non-associative

d. The bias vector becomes zero

Linear mappings are closed under composition: $\mathbf{W}_2(\mathbf{W}_1\mathbf{x}) = (\mathbf{W}_2\mathbf{W}_1)\mathbf{x} = \mathbf{W}_{eq}\mathbf{x}$.

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Section 2: Neural Parameter Counting

4 Calculate the total number of trainable parameters (weights + biases) in a dense layer with 256 inputs and 128 outputs.

a. 32928 ✓

b. 32768

c. 33024

d. 32896

Weights: $256 \times 128 = 32768$. Biases: 128. Total $= 32768 + 128 = 32928$.

5 A two-layer perceptron has architecture $10 \to 20 \to 1$. How many total parameters are in the network (including all biases)?

a. 241 ✓

b. 220

c. 200

d. 231

Layer 1: $10 \times 20 + 20 = 220$. Layer 2: $20 \times 1 + 1 = 21$. Total $= 220 + 21 = 241$.

6 A convolutional layer has 32 filters of size 3×3 across 3 input channels with a bias per filter. How many learnable parameters does it contain?

a. 896 ✓

b. 864

c. 288

d. 928

Weights: $32 \times (3 \times 3 \times 3) = 32 \times 27 = 864$. Biases: 32. Total $= 864 + 32 = 896$.

Section 3: Rectified Activations and Dying Neurons

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7 Evaluate the standard ReLU activation function $\text{ReLU}(z) = \max(0, z)$ and its derivative for $z = -3.5$.

a. $\text{ReLU}(-3.5) = 0, \quad \text{ReLU}'(-3.5) = 0$ ✓

b. $\text{ReLU}(-3.5) = -3.5, \quad \text{ReLU}'(-3.5) = 1$

c. $\text{ReLU}(-3.5) = 0, \quad \text{ReLU}'(-3.5) = 1$

d. $\text{ReLU}(-3.5) = 3.5, \quad \text{ReLU}'(-3.5) = -1$

For $z < 0$, $\text{ReLU}(z) = 0$ and its derivative is 0.

8 How does the Leaky ReLU activation $f(z) = \max(0.01z, z)$ mitigate the 'dying ReLU' problem during backpropagation?

a. **It provides a non-zero gradient (0.01) for negative inputs, preventing weights from freezing** ✓

b. It normalizes inputs to have zero mean and unit variance

c. It bounds outputs between -1 and 1

d. It eliminates the need for bias vectors

For $z < 0$, $f'(z) = 0.01 > 0$, allowing error gradients to flow back through inactive neurons.

9 Find the output of a Leaky ReLU activation with coefficient $\alpha = 0.1$ when input is $z = -8$.

a. -0.8 ✓

b. 0

c. -8.0

d. 0.8

$f(-8) = 0.1(-8) = -0.8$.

Section 4: Saturating Activations and Softmax Probabilities

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10 The standard logistic sigmoid is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.8$, compute the derivative $\sigma'(z_0)$.

a. 0.16 ✓

b. 0.80

c. 0.64

d. 0.20

$\sigma'(z) = \sigma(z)(1-\sigma(z))$. Here $\sigma'(z_0) = 0.8(1-0.8) = 0.8(0.2) = 0.16$.

11 Why does the Hyperbolic Tangent activation $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$ generally train faster than standard sigmoid in hidden layers?

a. **It is zero-centered with outputs in $(-1, 1)$, preventing systematic gradient bias** ✓

b. Its derivative is identically 1 everywhere

c. It does not require exponent calculation

d. It eliminates all negative activations

Being zero-centered means average outputs are near 0, preventing zig-zag gradient updates across positive coordinates.

12 Apply the Softmax function $\text{softmax}(\mathbf{z})_i = \frac{e^{z_i}}{\sum e^{z_j}}$ to the logit vector $\mathbf{z} = [0, \ln 2, \ln 5]^T$. Find the probability of class 3.

a. $\frac{5}{8} = 0.625$ ✓

b. $\frac{1}{8} = 0.125$

c. $\frac{2}{8} = 0.250$

d. $\frac{5}{7} \approx 0.714$

$e^0 = 1, e^{\ln 2} = 2, e^{\ln 5} = 5$. Sum $= 1 + 2 + 5 = 8$. $P(\text{class } 3) = 5/8 = 0.625$.

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